



Project Title: AI-Enabled Drought Early Warning and Climate-Resilient Decision Support System for Smallholder Farming: A Super-Hybrid Deep Learning Framework over the Choke Mountain Watershed, Ethiopia

Acronym: AI-DREWS

Team Composition & Expertise

Investigator Name & Qualifications	Project Role & Expertise Area	Institution	Core Responsibilities & Contributions
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Ambachow Kahsay (BSc)	Data Science & ML Engineering Lead	United States International University Africa	Deep learning architecture hyperparameter optimization, model training, and predictive pipeline engineering.
Bona Nasser (BSc, MSc)	Statistical Modeling & Climate Analytics Specialist	Haramaya University, Ethiopia	Model statistical benchmarking and performance evaluation.
Bethelhem Legesse (BSc)	Software Engineering & Dashboard/UX Developer	Raya University, Ethiopia	Front-end and back-end development of the decision-support platform
Eibrahim Belayneh (BSc)	GIS & Remote Sensing Specialist	Fusion IT Consultancy PLC	Spatial data collection and preprocessing (CHIRPS, ERA5, MODIS), vegetation condition mapping (NDVI/EVI), and spatial drought risk GIS modeling.

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Contents

1.	Introduction.....	3
1.1.	Novelty of the Proposed Project	4
1.2.	Specific Objectives	5
1.3.	Expected Outputs	6
2.	Data and methodology	7
2.1.	Data sources and resolution	7
	Table 1: data sources and resolutions	7
2.2.	Methodology	8
2.3.	Key steps: Super-hybrid CNN-LSTM-Fourier simulation model	9
3.	AI UniPod Timeline.....	10
	Table 2: Implementation Plan.....	10
	References	11

Executive Summary

AI-DREWS—AI-Enabled Drought Early Warning and Climate-Resilient Decision Support System for Smallholder Farming. It proposes a novel, high-impact AI-to-action platform that transforms global climate signals into localized, farm-level decisions for drought-prone smallholder agriculture in the Choke Mountain Watershed (CMW), Ethiopia. Current early-warning systems often stop at seasonal forecasts or risk maps; AI-DREWS goes further by creating a direct pathway from climate intelligence → AI prediction → localized drought risk → farm action. The project's core innovation is a Super-Hybrid CNN–LSTM–Fourier architecture that simultaneously learns spatial patterns, temporal dependencies, and periodic/extreme-event signals from 1990–2026 multi-source climate, satellite, hydrological, and agricultural datasets. These AI forecasts will be fused with drought indicators and remotely sensed vegetation conditions to generate 1–3-month localized drought-risk information. Critically, AI-DREWS will integrate Indigenous Ecological Knowledge (IEK) and farmer mental models to expand its prediction performance to 30 years, enabling the technology to move beyond technically accurate prediction toward locally trusted and usable decision support.

The project will produce three tightly connected innovations: (1) an AI climate engine capable of forecasting El Niño–Southern Oscillation (ENSO)-related hydro-climatic signals for drought/flood occurrences; (2) an AI drought-warning engine that converts these signals into spatially explicit drought probabilities; and (3) a decision-support Minimum Viable Product (MVP) that translates predictions into practical recommendations on crop selection, planting windows, and water management for crops. The proposed targets include ≥ 0.80 Niño 3.4 anomaly correlation at three-month lead, $\geq 10\%$ RMSE improvement over the best baseline, and validated drought-risk prediction performance.

Why AI UniPod should invest: AI-DREWS is designed not merely as a research model, but as a research-to-product innovation. Within six months, the project will progress from an AI-ready data cube to model development, drought-risk mapping, stakeholder validation, and a functional early-warning prototype. This creates a realistic pathway from research → prototype → field validation → scalable Ethiopian agri-tech solution, with the CMW serving as a living laboratory for future replication in other drought-prone farming systems. The funding opportunity is therefore catalytic: a relatively small AI UniPod investment can convert existing climate data, scientific expertise, and local knowledge into a demonstrable AI product with direct agricultural value. AI-DREWS will demonstrate how advanced AI can move from predicting climate risk to enabling farmers to act before drought impacts become irreversible.

Keywords: Predict, Localize, Act

1. Introduction

Climate variability is increasingly threatening sustainable crop production in the CMW, where irregular rainfall, recurrent drought, water scarcity, and climate extremes interact with agricultural and ecosystem vulnerabilities (Abate et al., 2025). Although farmers and institutions use seasonal forecasts and Indigenous ecological knowledge to guide agricultural decisions, existing forecasting approaches remain largely fragmented, retrospective, and insufficiently localized for timely farm-level action (Taklu et al., 2023). Most conventional systems treat climate variables and agricultural risks separately and provide information after a hazard has already developed. This creates a critical decision gap: there is no integrated, AI-driven early warning approach capable of translating large-scale climate signals such as El Niño and La Niña into localized, actionable drought-risk information for the CMW (Arash et al., 2025).

The proposed project introduces a novel AI-powered forecasting-to-decision framework that uses deep learning to learn the complex, nonlinear, and time-dependent relationships between El Niño–La Niña variability and drought development over the CMW (Arash et al., 2025). Rather than simply predicting drought from historical weather data, the project will investigate whether CNN–LSTM-based models can extract spatial and temporal climate signals from ENSO and related hydro-climatic datasets and provide earlier, more reliable drought warnings. This creates a direct pathway from global climate signals → AI forecasting → localized drought early warning → climate-resilient agricultural decisions. The innovation therefore lies not only in applying AI, but in transforming AI from a prediction tool into a practical decision-support system tailored to a climate-vulnerable watershed.

The project will further establish a foundation for co-producing AI-based climate services with local stakeholders, enabling forecasts to be translated into timely decisions on crop selection, planting periods, water management, and drought preparedness. This combination of deep learning, ENSO forecasting, localized early warning, and actionable agricultural decision support represents a significant step beyond conventional statistical forecasting and existing fragmented early-warning practices. The CMW will serve as a living laboratory to demonstrate how AI can turn complex climate information into usable knowledge for farmers, decision-makers, and institutions, with the potential to scale the approach to other drought-prone agricultural systems in Ethiopia.

1.1. Novelty of the Proposed Project

AI-DREWS introduces a novel, farmer-centered AI solution that transforms climate data into actionable drought decisions for agricultural risk reduction. Its key innovations are:

1. **Super-Hybrid AI Forecasting:** Unlike conventional statistical, dynamical climate, or single-model AI approaches, AI-DREWS integrates CNN, LSTM, and Fourier Transform to learn spatial patterns, temporal dependencies, and seasonal/extreme-event periodicity signals. This enables more effective representation of complex, nonlinear ENSO–drought relationships.
2. **Multi-Source Climate Intelligence:** The framework fuses ERA5, CHIRPS, NOAA (ENSO indices), satellite observations, and local meteorological data (1990–2026) to overcome the coarse resolution, computational burden, and systematic biases often associated with conventional GCM/RCM-based approaches, while enabling efficient real-time forecasting.
3. **AI + Indigenous Knowledge for Localized Decisions:** Rather than relying solely on data-driven predictions, AI-DREWS integrates Indigenous Ecological Knowledge and farmer mental models to contextualize AI outputs, strengthen trust, and improve local usability and adoption.
4. **From Drought Prediction to Farm Action:** Unlike conventional drought-risk maps that mainly indicate hazard levels, AI-DREWS transforms forecasts into actionable micro-advisories, including crop selection, optimal planting windows, and deficit-irrigation options for crops (teff, wheat, and maize).
5. **Research-to-Product and Scale-Up Pathway:** The project will convert the integrated AI framework into a functional Minimum Viable Product (MVP) delivered through a simple mobile/web interface. With model validation, explainability, bias assessment, and farmer co-validation, the prototype creates a clear pathway from research → prototype → field adoption → scalable agri-tech solution beyond the CMW.

Why It Is Competitive for AI UniPod

The project's strongest innovation is not simply applying AI to climate data; it is converting AI predictions into a usable agri-tech product. The proposed CNN–LSTM–Fourier framework integrates multi-source climate, satellite, and local observations to identify complex drought signals and provide 1–3-month early-warning information. Unlike conventional climate-risk products that mainly provide maps or probabilities, AI-DREWS delivers decision-ready information directly relevant to farmers and agricultural stakeholders.

Feasibility and Product Potential: The project is data-ready, technically feasible, and prototype-oriented. It will use established datasets such as ERA5, CHIRPS, ENSO indices, satellite vegetation indices, and Ethiopian Meteorological Institute observations (1990–2026). Within six months, the project aims to develop and validate a functional AI-DREWS prototype, providing a clear pathway from research → prototype → field validation → scalable agri-tech startup.

1.2. Specific Objectives

Objective	Scope	Measurable target
O1 · Forecast AI & Climate Engine	Build and train a super hybrid CNN–LSTM–Fourier model capturing spatial patterns, temporal dependencies, and periodic seasonal cycles of ENSO variability and hydro-climatic extremes over the CMW.	Niño 3.4 anomaly correlation ≥ 0.80 at 3-month lead; $\geq 10\%$ RMSE reduction vs. best baseline
O2 · Warn Local Risk Mapping Engine	Integrate ENSO predictions with CHIRPS, MODIS, ERA5, and station data to compute SPI/SPEI and VCI and project spatial drought probability maps 1–3 months ahead.	Brier Skill Score > 0.15 and ROC-AUC ≥ 0.75 for $P(\text{SPEI-3} \leq -1)$ at 1-month lead
O3 · Act & Deploy Agri-Decision Support MVP	Co-design a user-centred decision-support system that translates drought probabilities into farm guidance co-validated with farmers and agricultural experts.	$\geq 70\%$ advisory agreement with expert panel; task success $\geq 80\%$ in usability tests ($n \geq 40$)

1.3. Expected Outputs

1. **AI-Based ENSO Forecasting Model:** A validated CNN–LSTM–Fourier deep-learning model capable of identifying complex spatial–temporal relationships in El Niño–La Niña signals and generating improved seasonal climate forecasts for the CMW.
2. **Localized AI Drought Early Warning System:** A prototype AI-powered drought early warning and risk-mapping system that converts ENSO and hydro-climatic predictions into timely, localized drought-risk information for the CMW.
3. **Climate-Resilient Decision-Support Tool:** An accessible AI-based decision-support framework that translates drought forecasts into actionable recommendations for crop selection, planting timing, water management, and drought preparedness, supported by stakeholder validation and a pathway for scaling to other Ethiopian watersheds.

2. Data and methodology

2.1. Data sources and resolution

Table 1: data sources and resolutions

Source	Variables	Resolution	Temporal	Period	Role
NOAA/ERSST	SST, SST anomalies, Niño 3.4	$\sim 2^\circ \times 2^\circ$	Monthly	1990–2026	CNN spatial input
ERA5	Temperature, precipitation, pressure, wind, radiation, soil moisture, evaporation	0.25° (~ 31 km)	Hourly–monthly	1990–2026	Atmospheric bridge; PET for SPEI
CHIRPS	Rainfall/precipitation	0.05° (~ 5 km)	Daily–monthly	1990–2026	Downscaling target; SPI
MODIS/VIIRS	NDVI/EVI, vegetation condition, LST	~ 250 m–1 km	16-day/monthly	2000–2026	VCI; impact validation
EMI station data	Rainfall, temperature, met. variables	Point/station	Daily/monthly	1990–2026	Ground truth; bias correction
Crop/farm data	Crop type, yield, planting date, drought impacts	Farm/field/admin unit	Seasonal/annual	1990–2026	Decision-rule calibration
Hydrological data	River flow, water availability, soil moisture	Station/grid	Daily/monthly	1990–2026	Irrigation water budget
Sentinel-2 + SRTM ADDED	Multispectral reflectance; 30 m DEM, slope, aspect	10–30 m	5-day / static	2017–2026	Sub-km covariates; field crop masks
Crop	Crop Type / Species	Type of crops grown (e.g., teff, maize, wheat)	Ministry of Agriculture, Local agricultural offices, Surveys	Seasonal / Annual	Farm / Kebele / Watershed

2.2. Methodology

The system comprises three coupled engines that map one-to-one onto the objectives. All components share a single versioned data cube on a common monthly time axis (1990–2026), with strict temporal separation between training (1990–2015), validation (2016–2020), and independent test (2021–2026) periods to prevent information leakage across ENSO events.

Innovation pathway: 1990–2026 climate data → CNN–LSTM ENSO forecasting → localised drought prediction → risk mapping → agricultural action. Each objective makes a distinct, measurable contribution within this pathway (Fig. 1).

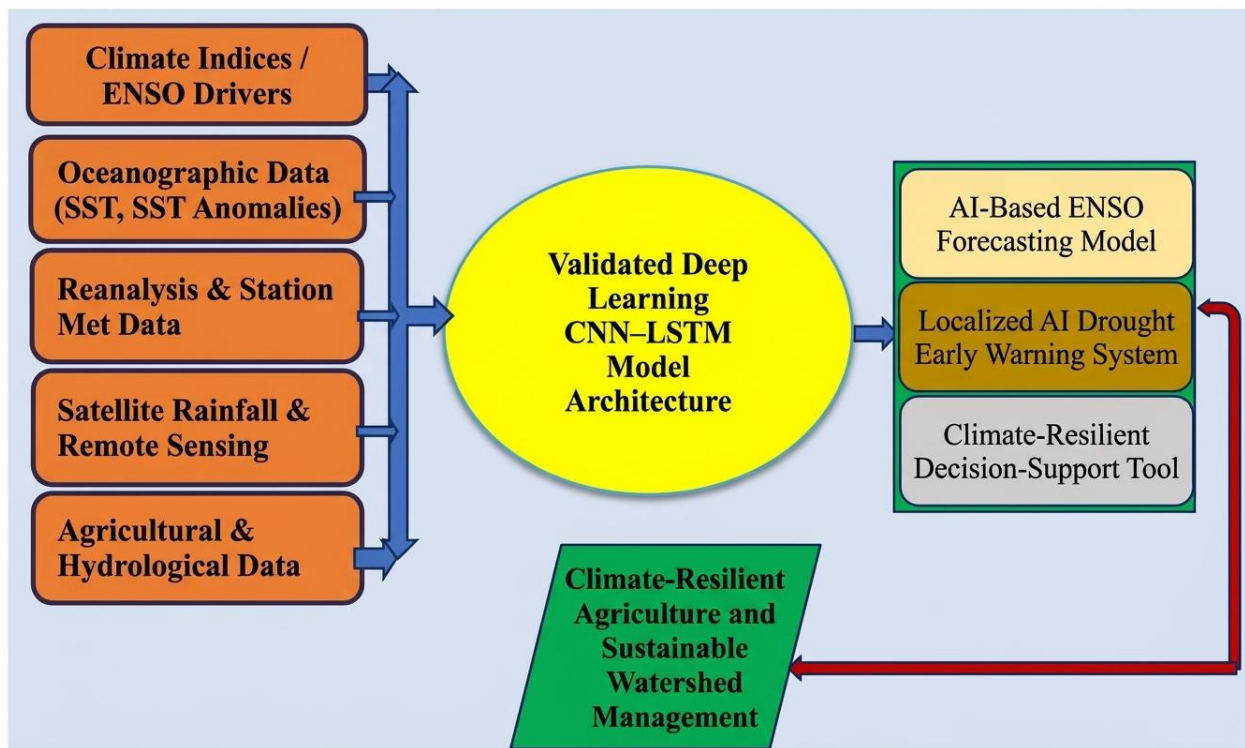


Fig. 1. Data structure for hybrid CNN-LSTM model drought forecasting.

The data will be split into training (70%) and validation & test 30% (15% and 15%) (Fig.2).

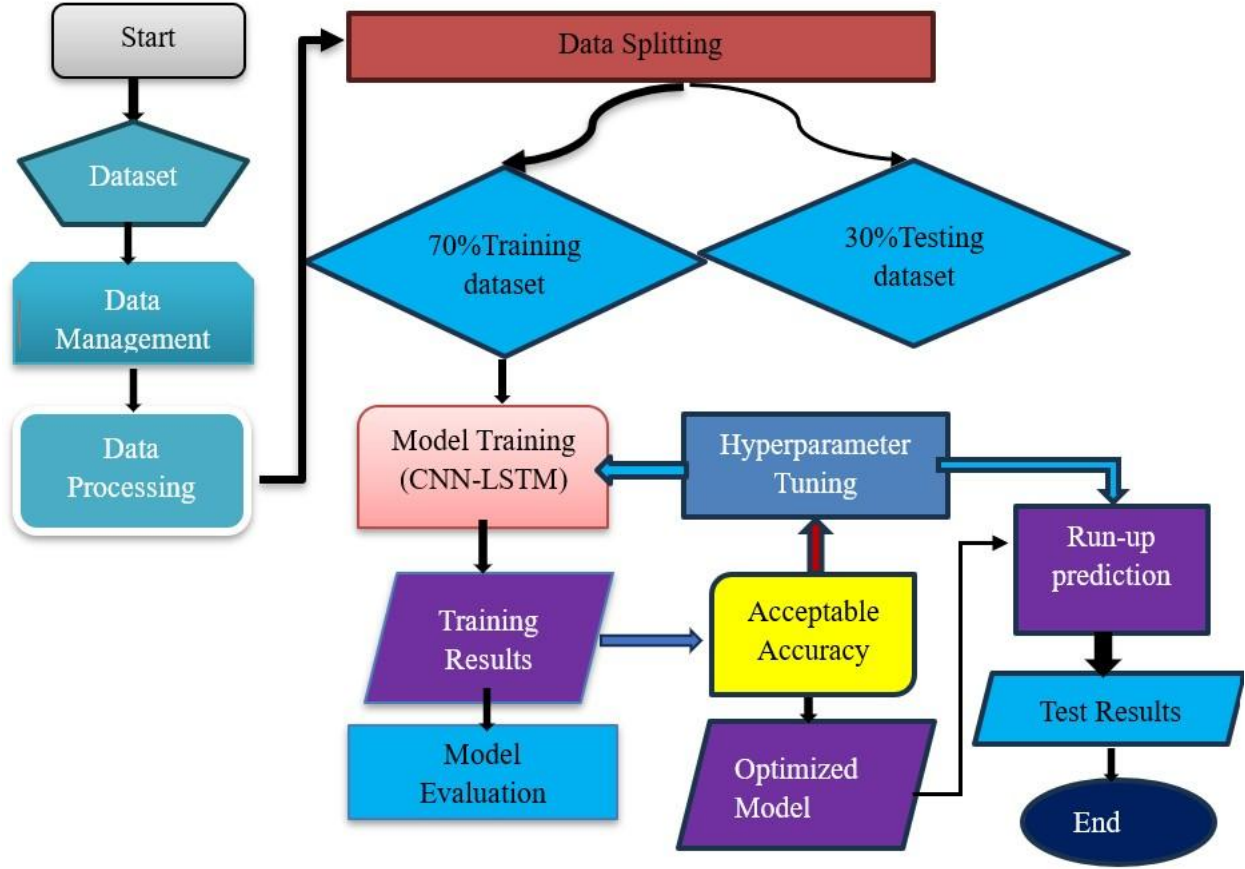


Fig.2. Data splitting and hybrid CNN-LSTM framework design.

2.3. Key steps: Super-hybrid CNN-LSTM-Fourier simulation model

To ensure realistic boundary and initial conditions, a super-hybrid CNN–LSTM–Fourier framework will be proposed for modeling crops under rainfed or irrigation at the CMW. This model integrates spatial, temporal, and periodic characteristics with agronomic water-balance principles. By utilizing multi-source datasets, it delivers real-time irrigation guidance while maintaining physical coherence and reliability.

The inputs of the spatial features (X_s): satellite indices (NDVI/EVI), soil maps, and crop spatial distribution.

The inputs of the temporal features (X_t): daily local weather/climate variables (temperature, precipitation, radiation, humidity, and wind).

The periodic signals (X_p): Fourier-transformed seasonal and extreme-event cycles (droughts, floods, heatwaves).

Testing against Indigenous Knowledge: The testing will involve comparing the scientific model's predictions (e.g., for drought/flood occurrence, rainfall patterns) with the forecasts derived from traditional indicators.

Mental modeling methods: These methods are designed to understand how individuals (in our case, farmers) perceive, interpret, and make decisions about complex systems like climate. This approach could help uncover the underlying logic and criteria farmers use in their indigenous forecasting.

3. AI UniPod Timeline

Table 2: Implementation Plan

Month	Key Activities	Expected Output
1	Collect, quality-control, and integrate 1990–2026 ENSO, ERA5, CHIRPS, satellite, and local data	AI-ready integrated database
2	Feature engineering and development of CNN–LSTM ENSO forecasting model	Preliminary AI forecasting model
3	Model optimization, training and independent validation	Validated ENSO forecasting model
4	Integrate ENSO forecasts with hydro-climatic indicators; develop AI drought prediction and risk mapping	Drought prediction model and risk maps
5	Validate drought forecasts and co-develop agricultural decision rules with stakeholders	Early-warning and decision-support framework
6	Develop, test and demonstrate the AI-powered early-warning prototype; dissemination and documentation	Functional AI drought early-warning & decision-support prototype

Month 1 → Data

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Months 2–3 → AI Forecasting

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Month 4 → Drought Early Warning

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Month 5 → Decision Support

↓

Month 6 → AI Prototype & Demonstration

Final 6-month outcome: A validated CNN–LSTM–Fourier ENSO forecasting model linked to a localized AI drought early-warning and climate-resilient agricultural decision-support prototype for the CMW climate change risk solution.

4. Indicative Budget

Table: USD are indicative and will be adjusted to the accelerator's funding ceiling.

Line item	Detail	USD	%
Personnel stipends	5 team members, 6 months, part-time	9,000	36
Fieldwork & participation	FGDs, co-design workshops, usability tests, transport, participant compensation	5,000	20
Compute & data	GPU cloud hours, storage, EMI station data fees	4,000	16
MVP delivery	SMS/USSD/IVR gateway, hosting, 10 test Android devices	3,500	14
Dissemination	Demo day, policy brief, publication fees	1,500	6
Contingency	Unforeseen field and data costs	2,000	8
Total		25,000	100

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